

Module 5

Troubleshooting Common Failures — Tier 1: Beginner

1. Warping

Corners of the print lift off the bed as it cools unevenly.

- Fix: better bed adhesion (clean bed, adhesive glue stick), enclosed printer, brim/raft in slicer

2. Stringing

Thin strands of plastic between parts of the print.

- Fix: lower nozzle temperature, enable/increase retraction settings, dry your filament

3. Poor First-Layer Adhesion

Print doesn't stick to the bed at all, or peels immediately.

- Fix: re-level the bed, clean with isopropyl alcohol, adjust first-layer height/squish

4. Layer Shifting

Layers suddenly offset mid-print, misaligning the model.

- Fix: check belt tension, reduce print speed, ensure printer is on a stable surface

5. Under-Extrusion / Gaps

Thin, weak, or gappy layers even though the file is correct.

- Fix: check for clogs, verify filament diameter setting, check flow rate calibration (Module 10)

General Troubleshooting Approach

- Change one variable at a time — never adjust multiple settings before re-testing
- Check the community — your exact printer model likely has documented fixes
- Keep a small log of what settings you changed and the result

